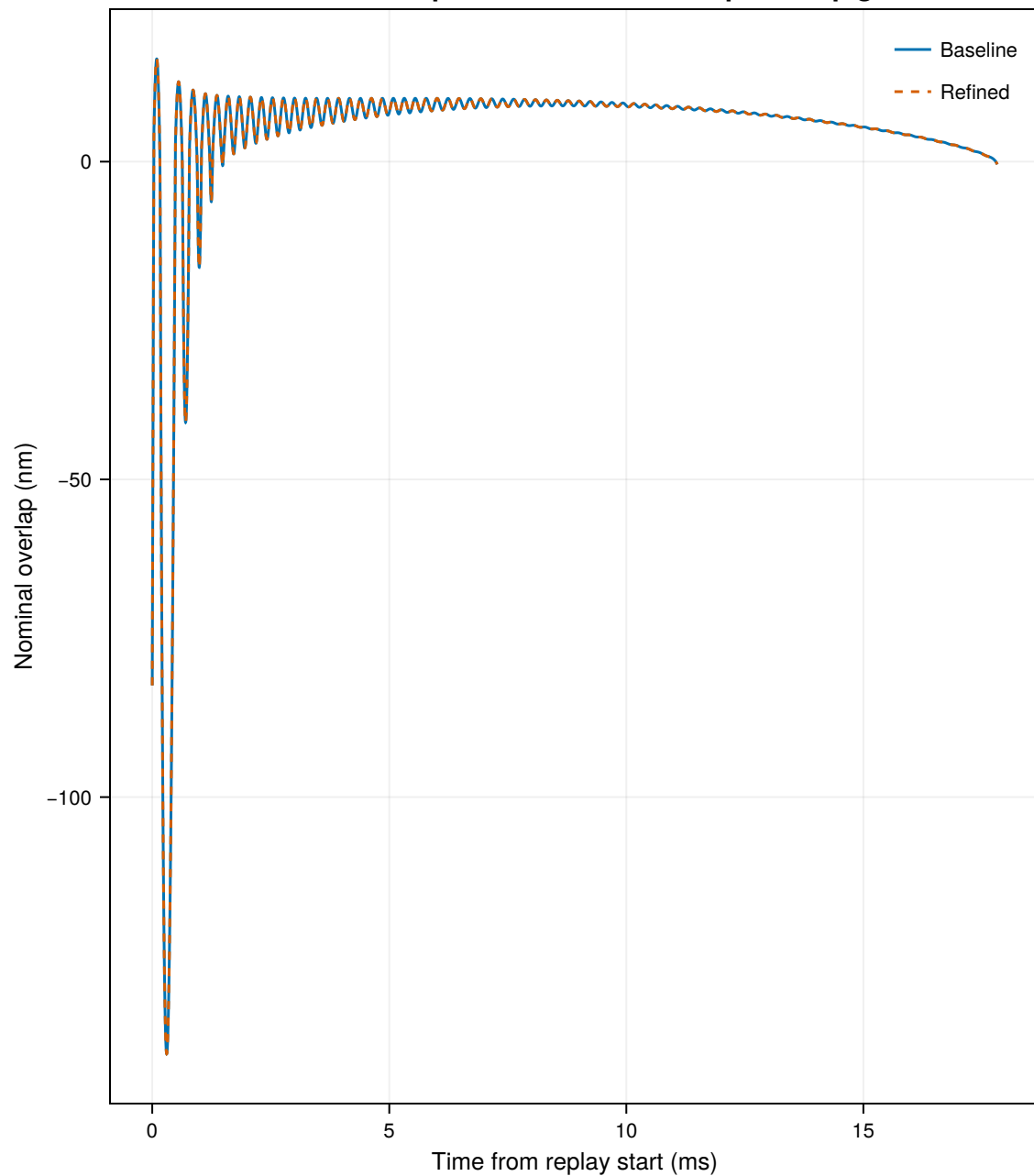


Full-sequence solver-refinement comparison
Side $r = -1$; all rebounds included; same initial state; tolerances / 5 and dtmax / 2

Contact overlap on the baseline accepted-step grid



Refined minus baseline at accepted-step times

